ASIC IMPLEMENTATION / ENGINEERING RESULTS

CNN Accelerator

Three-Round Physical Design Study

Trần Sơn Hiển

`CNN_300_cp3` → `CNN_adv` → `AMICA_R2_C16_ASIC`

Abstract

This report compares three retained implementations of a 1D CNN accelerator, focusing on SRAM organization, macro placement, floorplan constraints and post-route results. The progression is not a controlled same-architecture benchmark: clock targets, memory inventories and compute configurations differ. All three selected Innovus timing states meet their respective setup and hold constraints. Native geometry violations fall from 110 in the baseline to zero in the later rounds, but electrical-rule violations remain; the middle round also retains power-grid connectivity failures. The final AMICA implementation reaches a 250 MHz target with 54 SRAM macros and a 3.4998 mm^2 die. The evidence supports a portfolio of increasingly deliberate physical-design decisions, not a claim of workload-normalized performance improvement or foundry signoff.

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1 Scope and comparison rules

The study follows `CNN_300_cp3` (R1), `CNN_adv` (R2) and `AMICA_R2_C16_ASIC` (R3). Its purpose is to explain how memory organization and physical partitioning changed, show what each implementation achieved, and preserve the issues that remain open. It is a results report, not an RTL release or a reproduction of the earlier development diary.

Cadence Genus provides synthesis evidence; Cadence Innovus provides floorplan, placement, clock-tree, routing, timing, power and native physical checks. Native Cadence Tempus rechecks are reported separately where retained. The designs use a Nangate/FreePDK45-based academic flow and hard SRAM macros. These are not uniformly characterized foundry-signoff conditions.[E1–E6]

The selected R1 and R2 timing/power states are post-route and pre-filler; R3 is the final post-filler verification/export state. Later analysis-only rechecks do not constitute new physical implementations. Existing reports and saved databases were inspected read-only for this comparison; no new EDA optimization or closure run was performed.

How to read the numbers. Frequency is the constrained target, not measured throughput or characterized maximum frequency. WNS/TNS are conditional on each round's own clock and analysis settings. Power is vectorless, not workload-measured energy. SRAM bits describe physical macro inventory, not application-usable capacity. Geometry, signal connectivity, PG connectivity, electrical DRVs and independent signoff are separate acceptance domains.

2 Results across three rounds

Table 1: Selected timing states and native check results. Slack and TNS are in ns; counts are reported violations, not inferred zeros.

Metric	R1: baseline	R2: CNN_adv	R3: AMICA
Clock period / target	3.334 / 299.94 MHz	5.000 / 200 MHz	4.000 / 250 MHz
Selected physical state	Routed, pre-filler	Routed, pre-filler	Final, post-filler
Innovus setup WNS	+0.007	+0.033	+0.031
Innovus hold WNS	+0.001	+0.010	+0.010
Setup / hold TNS	0 / 0	0 / 0	0 / 0
Setup / hold violating paths	0 / 0	0 / 0	0 / 0
Max-capacitance violating nets	700	1042	797
Max-transition violating nets	0	0	2
Max-fanout violating nets	0	1	0
Native geometry DRC	110 (Metal9)	0	0
Regular-net connectivity	0 errors	0 errors	0 errors
PG-specific continuity	Not established	Failed; 1000 markers at report cap	Not established
Independent Calibre DRC/LVS	Not demonstrated	Not demonstrated	Not demonstrated

The timing rows use the final baseline post-route optimization, the selected `CNN_adv` post-route optimization, and AMICA's final Innovus recheck. They do not substitute the baseline's earlier negative routing result or mix native Tempus margins into an Innovus row. The PG failure

in R2 remains significant even though native geometry DRC and regular-net connectivity are clean.[E1,E3,E6]

Table 2: Geometry, SRAM inventory and power. Area is measured geometry or LEF-derived macro footprint; power stages differ explicitly.

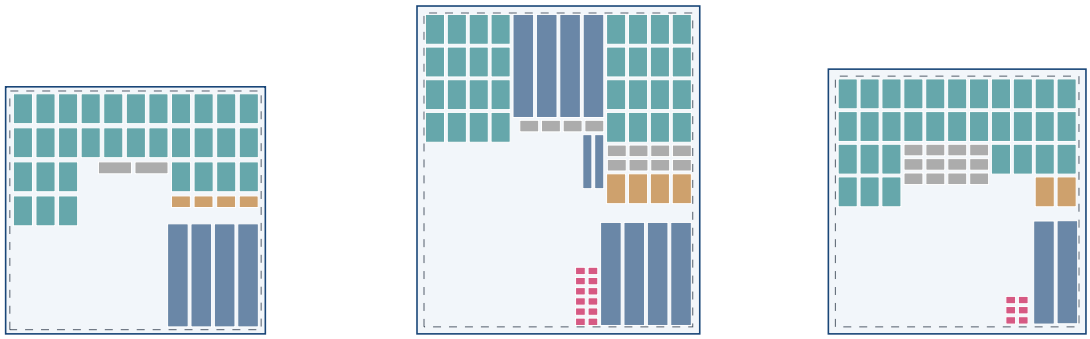
Metric	R1	R2	R3
Die area (mm^2)	3.2928	4.7574	3.4998
Core area (mm^2)	3.0787	4.3288	3.1346
SRAM macros	42	70	54
SRAM footprint (mm^2)	1.4582	2.1552	1.3703
Nominal SRAM inventory (KiB)	100.0000	144.3750	90.1875
Internal power (mW)	147.031	123.522	98.908
Switching power (mW)	60.293	26.684	41.079
Leakage power (mW)	6.972	7.342	5.212
Total estimated power (mW)	214.296	157.547	145.200
Power state	Pre-filler	Pre-filler	Post-filler

Power uses default/vectorless activity of 0.2 and the setup view, with a 0.95 V reported rail. The standard-cell and SRAM library corners are mixed rather than a uniform single-PVT characterization. Different targets, architectures, capacity and finishing stages prevent a like-for-like power-efficiency conclusion.[E1,E3,E6]

Relative to R2, R3 has a 26.4% smaller die and a 250 rather than 200 MHz target. These are descriptive changes, not a 25% throughput gain: retained synthesis RTL changes from a four-row, sixteen-column PE configuration to a two-row, sixteen-column wrapper. No common workload, end-to-end cycle count or equivalence regression is established here.[E4,E6,E7]

3 Floorplan evolution

Figure 1 reconstructs the SRAM geometry from the selected saved floorplans. The same physical scale is used for all rounds. This directly shows the larger memory envelope in R2 and the more compact arrangement in R3; it does not visualize standard-cell congestion, timing criticality or power-grid continuity.



(a) R1: 42 SRAMs **(b)** R2: 70 SRAMs **(c)** R3: 54 SRAMs

Feature maps Weights / sparsity Partial sums
Bias Instruction / descriptor Dashed line: core boundary

Figure 1: Data-derived macro maps from selected routed/final checkpoints, with matched LEF dimensions and stored orientations. These are measured-placement diagrams, not Innovus screenshots. Standard cells, routing, halos and module constraints are omitted for clarity. All 42/70/54 SRAM records are included.

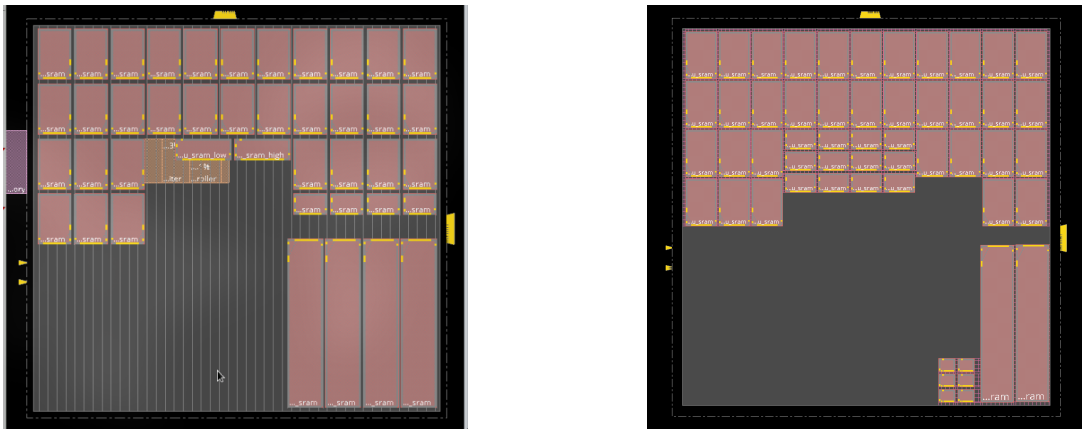
Table 3: Physical partitioning recorded in the selected saved floorplans. A guide is not a hard fence.

Decision / record	R1	R2	R3
Initial sizing policy	Fixed die, 30 μm margins	50% target, 50 μm margins	50% target, 50 μm margins
Macro-placement method	Explicit instance coordinates	Manual macro boxes and width refinement	Manual macro review gate
Hard fences / regions	0 / 4	3 / 4	0 / 0
Placement guides / soft guides	0 / 1	6 / 0	13 / 3
Macro halo	5 μm	5 μm	5 μm

R1: fixed memory geometry. Feature-map banks form a band, the long weight SRAMs occupy the lower portion, and the two instruction SRAMs sit centrally. Explicit placement establishes available channels before automatic standard-cell placement.[E1,E7]

R2: more memory and stronger partitioning. Feature-map SRAMs occupy the upper and side bands; weight and sparsity storage extend along the lower edge, with partial sums near the lower central area. Bias, descriptor and instruction storage occupy distinct groups. Hard regions/fences constrain central logic, but the final placement checks still report constraint violations.[E3,E7]

R3: a smaller memory mix and softer guidance. Feature-map banks are split to either side, instruction/descriptor SRAMs sit centrally, and weight/partial-sum storage occupies the lower-right portion. The saved final floorplan contains guides and soft guides, but no hard fence or region records. This is a change in constraint strategy, not proof that guides caused the final timing margin.[E5,E7]



(a) Baseline report: floorplan view **(b)** AMICA: macro-only view

Figure 2: Retained native-tool views as visual context. The baseline image belongs to the earlier project report. The AMICA image is historical and has no established final-checkpoint identity. Neither image is used to derive final timing, DRC counts or quantitative geometry.

4 Round 1: the routed baseline

`CNN_300_cp3` establishes the reference physical implementation. Its selected input is a mapped 300 MHz netlist; the saved clock period is 3.334 ns (approximately 299.94 MHz). Matching source RTL is not retained in this project folder, so the comparison uses its implementation evidence rather than assuming an RTL-to-RTL equivalence with later rounds.[E1,E2]

4.1 Floorplanning decisions

The baseline uses a fixed rectangular die with nominal 30 μm core margins. All 42 SRAM macros are placed by explicit instance coordinates, assigned orientations and fixed before standard-cell placement. The selected saved floorplan contains 38 macros at R0 and four at MX, with 5 μm halos. This makes macro geometry and the channels left for logic and routing explicit design decisions rather than incidental placer output.[E1]

The recovered checkpoint measures 1860.48×1769.88 μm. The scripted height is 1769.92 μm; the measured checkpoint geometry, not that nominal input, is used for area comparisons. SRAM occupies 1.4582 mm^2 of the 1.8251 mm^2 combined functional-cell and macro footprint. Memory geometry therefore dominates the floorplanning problem.[E2]

4.2 Closure sequence and remaining issues

The first routed reports are not the final result. After routing, setup WNS is -0.246 ns and hold WNS is -0.065 ns. Subsequent post-route optimization reaches $+0.007$ ns setup and $+0.001$ ns hold, with zero reported violating paths and zero TNS in both modes. The later restore/recheck evidence corroborates these margins.[E1]

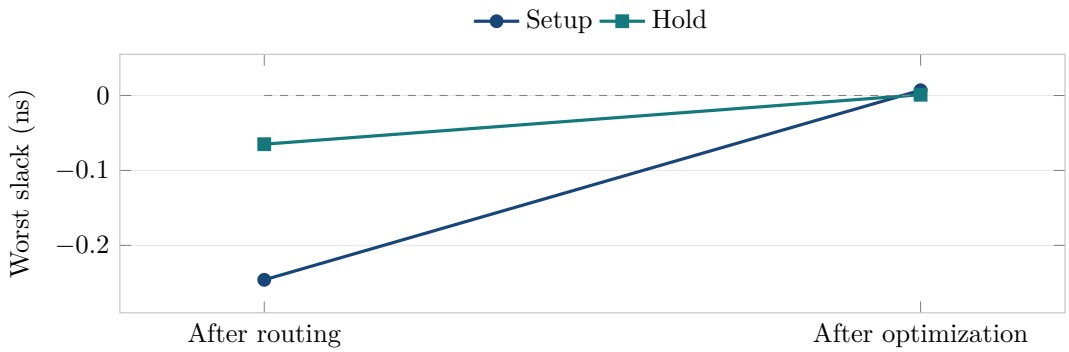


Figure 3: Round 1 timing before and after post-route optimization. These are two stages of the same baseline, not results from different CNN rounds.

Timing closure did not close every domain: 700 max-capacitance violating nets and 110 Metal9 spacing violations remained. Regular-net connectivity reported zero violations and zero warnings. The timing and power reports predate filler insertion. The later export includes fillers, but a separate post-filler timing closure result was not demonstrated.[E1]

Baseline outcome: a timing-met routed implementation with explicit macro placement and useful failure evidence, but not a DRC-clean or electrically closed design.

5 Round 2: CNN_adv

The intermediate design, `CNN_adv`, is routed and optimized at a 5.000 ns period (200 MHz). Later timing-only rechecks restore the same routed state and are not new optimization rounds.[E3]

5.1 What changed physically

The SRAM inventory rises to 70 macros: 36 of 16×1024, twelve of 16×16, twelve of 16×256, eight of 16×4096 and two of 4×2048. Wide logical memories are assembled from narrower physical SRAM leaves; logical wrapper count must not be substituted for physical macro count. The total is 144.375 KiB of nominal physical SRAM.[E3]

The initial sizing command uses a 50% target and 50 μm margins. Manual macro boxes, module constraints and a subsequent width reduction produce a 2025.59×2348.64 μm die. The saved final state has three fences, four regions and six guides. This is stronger hard partitioning than R3, not merely a larger version of the baseline.[E3,E7]

The retained synthesis source configures sixteen columns and four rows of PEs, while AMICA's wrapper selects sixteen columns and two rows. That architectural/configuration change is another reason not to interpret later frequency or power changes as physical optimization alone. These source-derived configuration counts do not establish workload performance or functional equivalence.[E4]

5.2 Timing met; other domains remained open

The final Innovus summary reports setup WNS +0.033 ns and hold WNS +0.010 ns, with zero TNS and zero violating paths. The later bounded native Tempus recheck reports +0.050 ns and +0.008 ns after fresh low-effort RC extraction; an earlier bare Tempus invocation reports

+0.001 ns and +0.002 ns. These are different analysis configurations and are not interchangeable margins.[E3]

Native geometry DRC and regular-net connectivity report zero errors. The PG-specific report, however, contains four disconnected special-wire problems and 996 dangling-wire markers before reaching the 1000-marker reporting cap. Placement checks also report 13,827 region/fence violations and 2,493 not-of-fence violations; electrical checks retain 1042 max-capacitance nets and one max-fanout net.[E3]

The selected project retains its saved implementation and reports, but no completed final DEF/SPEF delivery bundle. A separate verification stream-out GDS is not the same as a verified final export package. Current flow scripts were edited after the selected run, so the executed historical logs are the provenance for its result, not those newer scripts.[E3]

Intermediate outcome: a larger, explicitly partitioned, timing-met 70-macro implementation with clean native geometry DRC, but unresolved PG continuity, electrical and placement-constraint issues. It is not a signoff-clean result.

6 Round 3: AMICA physical implementation

The final round is `AMICA_R2_C16_ASIC`. Its canonical synthesis and physical constraints use a 4.000 ns clock period, corresponding to a 250 MHz target. The selected final export contains the routed netlist, DEF, GDS and RC-best/RC-worst SPEF files. Export completion establishes the availability of those implementation artifacts, not tapeout qualification.[E5,E6]

6.1 Memory-heavy placement and physical partitioning

The final DEF contains 54 SRAM macros across four types: 34 instances of 16×1024, 12 of 16×256, six of 16×16, and two of 16×4096. Their name-implied physical inventory is 738,816 bits (90.1875 KiB); this is not an assertion of application-usable capacity.[E6]

The floorplan script uses a utilization-based initial sizing policy with a 0.5 target and 50 μm margins, followed by an explicit manual macro-placement review gate. The selected saved final floorplan contains thirteen guides and three soft guides, with no hard fence or region records. It records guidance for the processing-element array, controller, instruction and descriptor memories, line buffer, arbiters, partial-sum memory and other blocks. These guides document physical partitioning intent; they do not by themselves prove improved congestion or shorter critical paths.[E5]

The final die measures 1845.47×1896.44 μm ($3.4998\,\mathrm{mm}^2$). The functional standard-cell plus SRAM footprint is $1.6498\,\mathrm{mm}^2$, excluding fillers and welltaps. Although the macro count exceeds Round 1, the SRAM footprint is smaller because the macro mix differs. Macro count alone is therefore not a measure of memory capacity or occupied area.[E6]

6.2 Final checks and the boundary of closure

The separate final Innovus timing recheck reports setup WNS +0.031 ns and hold WNS +0.010 ns. Native Tempus analysis reports +0.035 ns and +0.009 ns respectively. Both modes report zero TNS and zero violating paths. The engines are retained as separate results rather than choosing the more favorable number.[E6]

The final Innovus geometry check and regular-net connectivity check each report zero violations. However, 797 max-capacitance violating nets and two max-transition violating nets remain in the Innovus setup summary. Positive timing margins and zero geometry errors do not erase those electrical violations. A dedicated clean PG-continuity result is not established by the selected regular-net check.[E6]

Final extraction is recorded as low-effort and uncoupled, and the RC corners use a single typical capacitance table. The standard-cell and SRAM libraries also have mixed voltage/temperature characterization. These limits prevent a full RC/PVT/signoff claim, even though the selected timing views meet their constraints.[E6]

Final-round outcome: a 54-macro routed implementation exported at a 250 MHz target, with positive reported setup/hold margins and clean native geometry and regular-net connectivity checks. Electrical-rule closure, qualified corners, SI verification and independent Calibre verification remain separate work.

7 Lessons and next checks

Macro organization is a primary physical-design variable. The three rounds use 42, 70 and 54 SRAM macros with different widths, depths and aspect ratios. A larger macro count does not necessarily mean a larger SRAM footprint: R3 has more macros than R1, yet less macro area. Floorplan comparisons should therefore include the macro mix and real dimensions, not just counts.

Constraints should express intent without being mistaken for closure. The progression combines absolute macro coordinates, hard regions/fences, and later guide-based partitioning. The unresolved placement-constraint checks in R2 show why creating a fence is not sufficient; the final placement must be checked against it. Congestion improvement cannot be inferred merely from a cleaner-looking macro map.

Stage identity matters. The baseline's negative first-route timing is superseded by a later timing-met post-route result. Conversely, pre-filler timing must not be presented as a demonstrated post-filler recheck. Final implementation logs, checkpoint lineage and report generation stages should agree before a metric is promoted into a comparison.

A credible result retains its limits. Positive setup/hold slack coexists with electrical violations in all rounds. Zero native geometry DRC coexists with failed PG continuity in R2. Native tool checks are not independent Calibre verification.

7.1 Remaining verification work

1. Close max-capacitance, transition and fanout violations, then rerun timing at the same selected physical state.
2. Repair and recheck PG continuity separately from regular-net connectivity; resolve hard placement-constraint violations where present.
3. Qualify consistent library/RC corners, coupling and SI settings; retain full timing coverage and detailed diagnostics.
4. Run independent Calibre DRC and LVS with a matched technology/rule setup. Calibre closure is future work, not an achievement claimed by this report.

5. Establish a common functional regression and workload before publishing throughput, energy per inference or architecture-normalized efficiency comparisons.

Conclusion. The portfolio demonstrates three distinct physical implementations and an evolution in SRAM-aware floorplanning and partitioning. The final round delivers a 54-macro, 250 MHz-target routed/exported design with positive reported setup/hold margins and clean native geometry/regular-net checks. Remaining electrical, PG, model and independent-verification boundaries are preserved rather than hidden.

Evidence notes

The numbered markers identify retained engineering evidence, not external publications. Exact file mappings, hashes, source lines and consolidation corrections are maintained in a separate private audit record; raw implementation databases and technology files are not part of the report source package.

E1 Baseline implementation commands and parent logs; final post-route optimization, restore/recheck timing, native geometry/connectivity and vectorless power reports.

E2 Baseline saved geometry and class-filtered functional-cell/SRAM footprint audit; physical-only finishing cells are excluded from the quoted active footprint.

E3 `CNN_adv` selected routed checkpoint, final parent-log timing, power, placement, geometry, regular/PG connectivity, SRAM leaf inventory and separately retained native Tempus rechecks.

E4 Retained `CNN_adv` synthesis-top parameters and AMICA fixed-geometry wrapper. Source configuration is not a workload or equivalence test.

E5 AMICA implementation inputs and saved placement intent, including its manual macro-review gate and guide-based partitioning.

E6 AMICA final exported checkpoint and DEF inventory; final Innovus/Tempus reports, post-filler vectorless power, native checks and extraction/model qualifications.

E7 Selected saved floorplan block/constraint records and matched SRAM LEF dimensions for all three rounds. Arithmetic, full macro counts and plotted geometry were checked independently during consolidation.